KING KHALID UNIVERSITY ■ College of Computer Science

Genetic Mutation Detection Using Machine Learning

Presented by

Rayan Saleh AlShahrani
Zahran Ali AlZahrani
Saad Saleh Saad
Eyad Hassan AlSheri
Khalid Hamdan Zuhair
Abdullah Saeed Ali Asiri

Supervised by

Mr. Makki Akasha Babiker Al-Bashir

Problem statement: missense variants at clinical scale

Missense variants at clinical scale

- A point mutation that swaps one amino acid in a protein.
- Most common coding variant in the human genome.
- Most are benign; a clinically-important minority cause disease.
- Distinguishing the two is the central problem of clinical genetics.

Why prior work overstates performance

1. **Definitional circularity** — features derived from the label (e.g. `is_common`) inflate metrics.
2. **Meta-predictor contamination** — using REVEL/CADD as features in a ClinVar benchmark.
3. **Paralog leakage** — KRT1 in train + KRT14 in test $\Rightarrow$ 80% sequence identity $\Rightarrow$ model has seen most of the test.

Exome: $\sim$20,000 variants/patient; human curation $\sim$a few hundred/day. *Scale demands automation; automation demands honest evaluation.*

Research objectives + scope

Data + methodology

1. Assemble a paralog-disjoint corpus of 195k missense variants.
2. Eliminate three specific leakage sources.
3. Train and calibrate a defensible baseline.
4. Re-score published tools on our own test split.

Validation + delivery

5. Evaluate externally on DENOVO-DB.
6. Prove orthogonality of zero-shot ESM-2 signal.
7. Publish everything: code, tests, Docker, 10 narrative notebooks, Streamlit demo, academic report.

Every objective maps to at least one CI-enforced check.

Classical tools (2003–2010)

Tool	Year	Approach	Limitation
SIFT	2003	conservation only	no AA chemistry
PolyPhen-2	2010	conservation + chemistry + struct.	HumDiv overlaps ClinVar

- Access: Ensembl VEP REST, instant.
- Coverage: ∼93–96% of our test variants.
- Threshold design: SIFT picks 0.05 arbitrarily.

Gradient-boosted ensembles (2016–2024)

Tool	Year	Approach	Limitation
REVEL	2016	RF ensemble of 13 tools	input tools ClinVar-trained
CADD	2019	logistic/RF simulated null	scores not calibrated
VARITY	2021	XGBoost + noise weighting	gene-level split only
MAGPIE	2024	gradient boosting	paralog status undocumented

- All sit above 0.92 ROC-AUC on their own benchmarks.
- None report a *like-for-like* external test.
- None publish bootstrap CIs on every metric.

Protein language models (2021–2023)

Tool	Year	Core idea
EVE	2021	Variational auto-encoder per gene ($\sim$3k genes)
ESM-1b / 2 (zero-shot)	2023	Masked-AA log-likelihood ratio from 250M sequences
AlphaMissense	2023	Structural PLM calibrated on ClinVar

Why the PLM wave matters

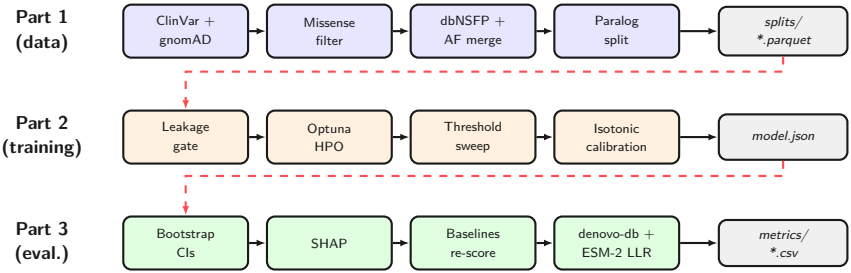
ESM-2 is fully unsupervised $\Rightarrow$ provably **uncontaminated** by ClinVar. Our Phase 2.1 integrates it as a training-time feature.

What we contribute

1. **Transparent leakage audit** with before-and-after numbers (PR-AUC 0.955 $\rightarrow$ 0.836).
2. **Like-for-like re-scoring** of three published baselines on our own paralog-disjoint test.
3. **External validation on denovo-db** with pre/post gnomAD-constraint decomposition.
4. **Two-pass ESM-2 integration** — Phase-1 zero-shot rank-fusion (ΔROC $+0.003$ test, $+0.015$ external) plus Phase-2.1 training-time LLR feature (ΔPR-AUC $+0.031$, $p < 10^{-4}$).
5. **Auditable methodological package**: 95% bootstrap CI on every metric, Murphy-decomposed calibration, paired bootstrap on ablations.
6. **Engineering artifacts**: 157 tests, CI-enforced leakage gate, Docker, Streamlit demo, LaTeX report.

`github.com/RayanAlDwlah/Genetic-Mutation-Detection-project`

Pipeline – three parts



Part 1 – Data preprocessing (1/3)

Sources:

- ClinVar – pathogenic / benign labels (review star ≥ 2)
- gnomAD v2.1 + v4 – allele frequency, AC, AN
- dbNSFP 5.3 – conservation + AA chemistry + Pfam
- gnomAD v2.1.1 – gene-level constraint (pLI, LOEUF, mis_z)

Join key: canonical `chr:pos:ref:alt` on GRCh38.

Part 1 – Data preprocessing (2/3)

The critical filter

Missense-only

Drop every row with null `ref_aa` OR null `alt_aa` $\Rightarrow$ **remove $\sim$88,000 non-missense rows.**

- Pre-audit: 64% of pathogenic rows had null AA fields!
- Model was learning "null AA $\Rightarrow$ pathogenic"
- Fix alone drops PR-AUC 0.955 $\rightarrow$ 0.819 (the single biggest audit finding)

Part 1 – Data preprocessing (3/3): paralog-aware split

- Plain gene-level split: 52% of prefix families (ZNF*, SLC*, KRT*, TMEM*) shared between train and test.
- Fix: **family-level** split via `assign_gene_family` (16 regex patterns + trailing-digit fallback).
- Result: 15,479 genes → 7,851 families, **zero** shared across splits, enforced by CI.

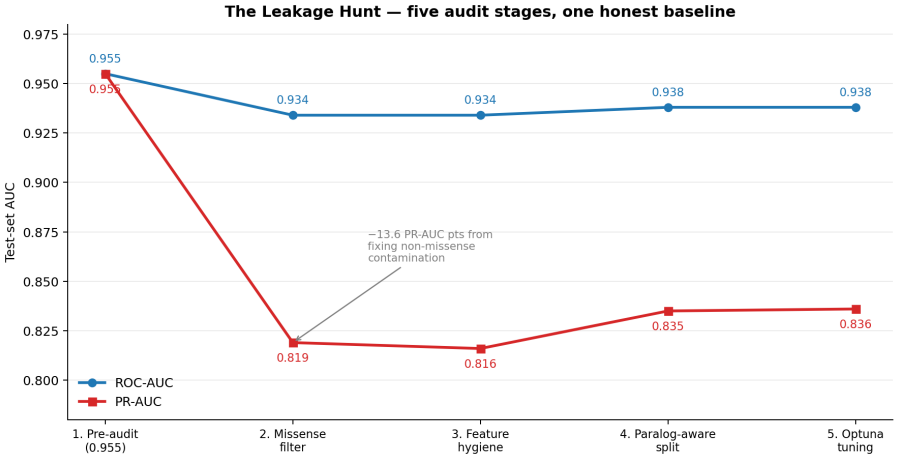
Part 2 – Leakage audit (1/4)

Four CI-enforced invariants

1. No banned features (`is_common`, `chr`, `ref`, `alt`) in the training matrix.
2. 100% missense-only rows in training split.
3. Zero shared gene families across train/val/test.
4. Pathogenic-rate gap < 0.08 across splits.

Every push to `main` runs the four checks.
A failure blocks the merge.

Part 2 – Five-stage leakage journey (2/4)



Part 2 – Model + calibration (3/4)

Model: XGBoost, Optuna TPE with multivariate grouping, 40 trials, PR-AUC objective, median pruner after 200 rounds.

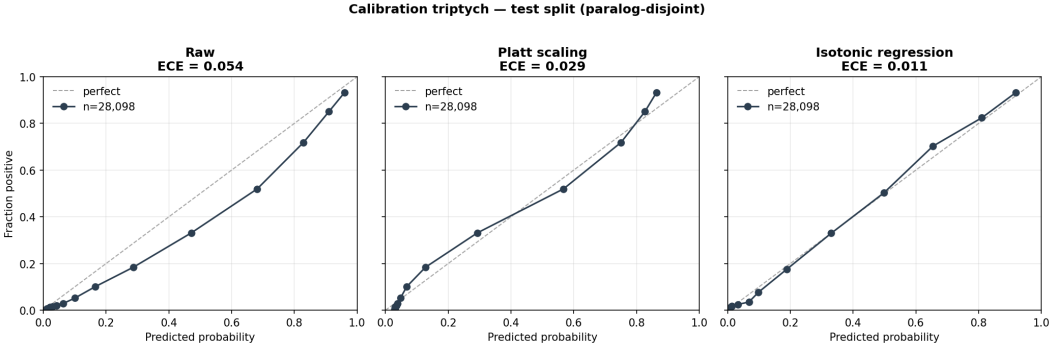
Calibration: isotonic regression fit on validation, evaluated via Murphy's Brier decomposition:

$$\text{Brier} = \text{Reliability} - \text{Resolution} + \text{Uncertainty}$$

	Brier	Reliability	Resolution	ECE
Raw	0.0876	0.00543	0.1055	0.054
Platt	0.0834	0.00114	0.1055	0.029
Isotonic	**0.0826**	**0.00024**	0.1053	**0.011**

- Resolution constant $\Rightarrow$ discrimination unchanged.
- Reliability drops 23$\times$ $\Rightarrow$ calibration works.

Part 2 – Reliability triptych (4/4)



Part 3 – Headline results: paralog-disjoint test (1/3)

Test split
Paralog-disjoint
28,098 variants
Pathogenic rate: $\sim$30%

All confidence intervals from 1,000-replicate nonparametric bootstrap.

Metric	Value (95% CI)
ROC-AUC	0.938 [0.935, 0.941]
PR-AUC (uncalibrated)	0.836 [0.827, 0.843]
PR-AUC (calibrated)	**0.827** [0.819, 0.835]
F1	0.775
Brier (calibrated)	0.083
ECE (calibrated)	0.011

Calibration trades a small ranking penalty (ΔPR-AUC $= -0.008$, $0.8355 \rightarrow 0.8273$) for $\sim 5\times$ better probability reliability (ECE $0.054 \rightarrow 0.011$). Calibrated probabilities are the headline; uncalibrated is shown for transparency.

Part 3 – SHAP interpretability (2/3)

Highlighted rows: gene-level constraint features added in Phase 2.1.

Rank	Feature	mean |SHAP|
1	phyloP100way_vertebrate	0.809
2	AN (gnomAD allele number)	0.532
3	**lof_z (gnomAD constraint)**	0.342
4	alt_aa_P (Proline)	0.304
5	pfam_domain	0.284
8	**oe_mis_upper (gnomAD constraint)**	0.189

- Two Phase 2.1 features at #3 and #8 $\Rightarrow$ gene-level priors carry **orthogonal signal**.
- 326 of 2,000 test variants are confident errors (16.3%).
- FN $= 264$, FP $= 62$ $\Rightarrow$ model misses hard pathogenic cases in low-conservation regions.

Part 3 – External validation: denovo-db (3/3)

Highlighted rows: holdout from gene families completely unseen in training.

Slice	n	ROC-AUC (95% CI)	PR-AUC (95% CI)
full (pre-constraint)	642	0.468 [0.42, 0.52]	0.761
full (post-constraint)	642	**0.511** [0.46, 0.56]	**0.790**
holdout (pre-constraint)	201	0.487 [0.38, 0.58]	0.789
holdout (post-constraint)	201	**0.573** [0.48, 0.67]	**0.838**

The headline

On the family-holdout slice ($n = 201$): paired bootstrap ΔROC $= +0.076$ $[-0.021, +0.172]$, one-sided $p = 0.073$. Lower bound below zero — we frame this as **partial closure** of the generalisation gap, not a deployable predictor.

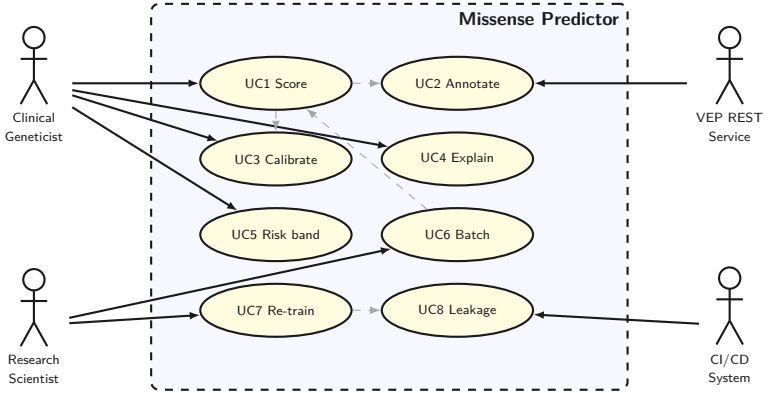
Functional + non-functional requirements

Functional ($\times 8$): ingestion, feature assembly, scoring, risk categorisation, feature attribution, batch mode, external reproduction, report regeneration.

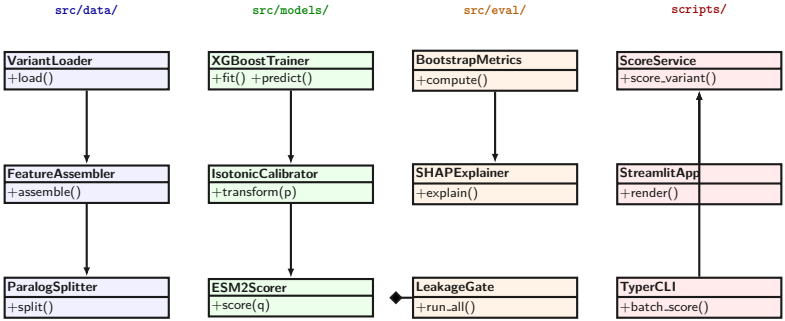
Non-functional ($\times 7$): performance ($<$100 ms cache hit), reliability (seed-reproducible to 10^{-3}), scalability (10^6 variants), usability (non-ML end user), maintainability (CI enforced), security (no PHI), reproducibility (Docker-replayable).

Traceability: every requirement maps to at least one `src/` module and at least one pytest test case.

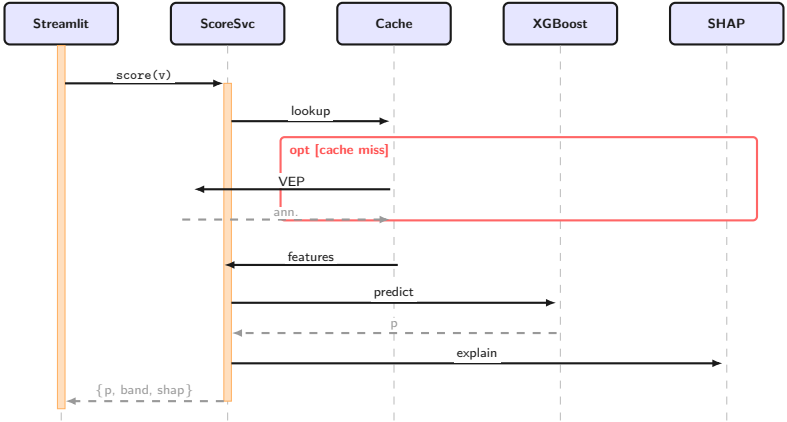
UML 1/6: Use case diagram



UML 2/6: Class diagram (grouped by package)



UML 3/6: Sequence diagram (scoring request)



System architecture (6-layer)

L6 Presentation – Streamlit ▪ Typer CLI ▪ Python API

L5 Application – validation ▪ cache ▪ payload

L4 Service – score_variant() ▪ explain() ▪ batch_score()

L3 ML core – XGBoost ▪ isotonic ▪ SHAP ▪ ESM-2 LLR

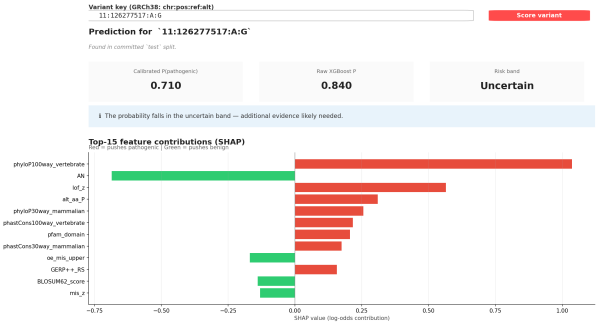
L2 Data access – pyarrow ▪ paralog splitter ▪ assembler

L1 Data – ClinVar ▪ gnomAD ▪ dbNSFP ▪ denovo-db ▪ UniProt

Quality
gates
`ruff`
`mypy`
`pytest`
leakage
CI

Repro-
ducibility
Docker
`uv.lock`
fixed seed
snapshots

Streamlit demo – scoring a variant



Input	GRCh38 variant key `chr:pos:ref:alt`
Output	calibrated p + raw XGBoost + risk band
Explainer	Top-15 SHAP, color-coded by direction
Latency	< 100 ms cache hit; ~ 2 s cache miss

Testing strategy: 5-level pyramid

1. **Unit** – 120 tests, <10 s
   pure functions, metric math, paralog regex
2. **Property** – 8 `hypothesis` properties
   calibrator monotonicity, rank preservation
3. **Integration** – 25 tests, 1–2 min
   leakage gate, cache fallback, module flows
4. **Scientific / E2E** – 5 scripts, 5–15 min
   reproduce-headline within 10^{-3}
5. **CI gate** – blocks merge on red

Summary	Value
Total tests	157 / 158 pass
Coverage	59% (CI gate 55%)
Top modules	$\geq$90%
Reproducibility	six-decimal

Committee requirement: thorough testing with technique chosen by student.

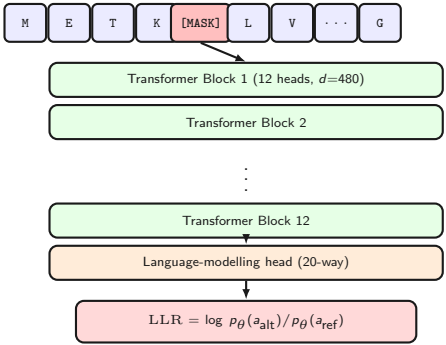
Leakage gate – CI-enforced invariants

```
$ python -m src.verify_no_leakage
```

```
============================================================
LEAKAGE GATE  --  src/verify_no_leakage.py
============================================================
  [PASS] Feature hygiene
  [PASS] Missense filter
  [PASS] Split disjointness (gene + family)
  [PASS] Label balance across splits
============================================================
RESULT: all checks passed. Baseline is leak-free.
```

All 4 invariants fail-closed: any red row **blocks the merge** to `main` via GitHub Actions.

ESM-2 zero-shot – scoring architecture



Model:

`facebook/esm2_t12_35M_UR50D`

- 12 transformer blocks
- $d = 480$, 20 attention heads
- $\sim$35M parameters
- **Zero-shot**: no weight updates

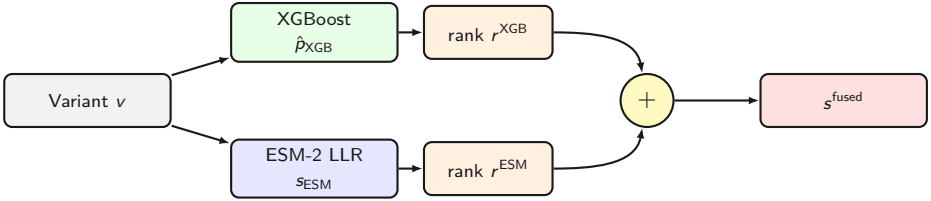
Log-likelihood ratio:

$$\mathrm{LLR}_{\text{ESM-2}} = \log \frac{\mathbb{P}_\theta(a_{\mathrm{alt}} \mid s_{\setminus\pi})}{\mathbb{P}_\theta(a_{\mathrm{ref}} \mid s_{\setminus\pi})}$$

where $s_{\setminus\pi}$ is the sequence with position π masked.

Lower LLR $\Rightarrow$ more pathogenic.

Rank fusion – operator + formula



$$s_i^{\text{fused}} = (1-\alpha)\cdot\frac{r_i^{\text{XGB}}}{n} + \alpha\cdot\frac{r_i^{\text{ESM}}}{n}, \qquad \alpha^* = 0.175$$

Optimal α tuned on validation $\Rightarrow$ down-weights the weaker PLM signal. Uniform α=0.5 is actually *worse* than XGBoost alone.

ESM-2 zero-shot + rank fusion

Test split (28,098 variants, paralog-disjoint)

Model	ROC-AUC	PR-AUC	α
XGBoost (calibrated)	0.938	0.827	—
ESM-2 LLR (zero-shot)	0.739	0.610	—
Rank fusion, uniform	0.895	0.784	0.50
Rank fusion, tuned	**0.941**	**0.852**	**0.175**

External (denovo-db, family-holdout):

- XGBoost: ROC 0.573
- ESM-2 alone: ROC 0.552
- Fusion: ROC **0.588** (+0.015)

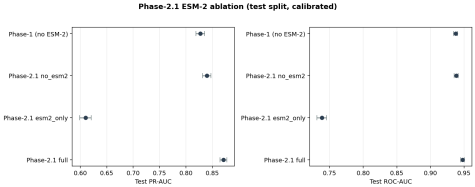
Two signals of orthogonality

Fusion improves over either source AND the gain transfers to held-out gene families. Motivates training-time integration of the LLR.

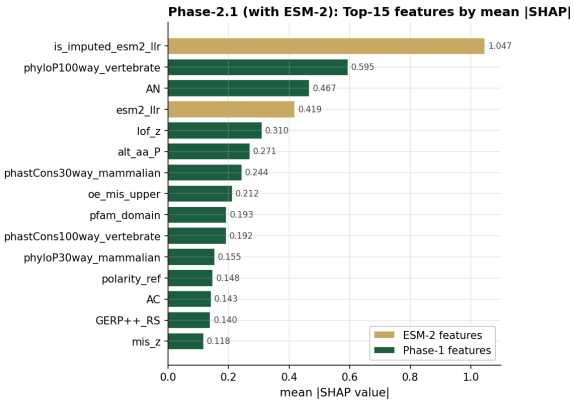
Training-time ESM-2 LLR integration: ablation

Model	Test ROC	Test PR
Baseline (no ESM-2)	0.938	0.827
Augmented `no_esm2`	0.939	0.840
Augmented `esm2_only`	0.739	0.610
Augmented `full`	**0.948**	**0.865**

- Paired bootstrap full vs no_esm2 (n=28,098):
- ΔPR-AUC $= +0.031$ $[+0.027, +0.036]$, $p < 10^{-4}$
- ΔROC-AUC $= +0.009$ $[+0.008, +0.010]$, $p < 10^{-4}$
- Frozen baseline hyperparams; Optuna re-tune $|\Delta| < 0.005$ (appendix only).



What ESM-2 changed in SHAP



Top features after ESM-2 integration:

- `is_imputed_esm2_llr` ranks #1 — the imputation flag itself carries strong signal (skipped variants live in non-canonical or truncated transcripts).
- `esm2_llr` ranks #4 — substantial but not dominant.
- Conservation features (phyloP, GERP) remain top-tier; ESM-2 partly substitutes, partly supplements (Spearman $\rho \leq |0.31|$).
- denovo-db holdout: ΔROC $= -0.13$ — in-distribution gain does not transfer at the 35 M scale.

Phase-1 top features were `phyloP100way_vertebrate`, `AN`, `lof_z` *(ranks 1, 2, 3); after ESM-2,* `lof_z` *drops to rank 5 while* `is_imputed_esm2_llr` + `esm2_llr` *take ranks 1 and 4.*

Comparison with published methods

All baselines re-scored on our paralog-disjoint test split (28,098 variants).

Method	Year	Methodology	Coverage	PR-AUC
SIFT	2003	legacy, no guards	96%	0.620
PolyPhen-2†	2010	legacy, no guards	93%	0.728
AlphaMissense*	2023	high ClinVar risk	86%	0.890
ESM-2 LLR (zero-shot)	2023	no supervision	100%	0.604
XGBoost (ours, baseline)	2026	paralog + audit	100%	0.827
XGBoost + ESM-2 LLR (ours)	**2026**	**paralog + audit**	**100%**	**0.865**

Each row reported on that baseline's own-coverage subset, so rows are NOT directly comparable; matched-coverage analysis in the thesis (Section 5.5) closes most of the AM gap.

* AlphaMissense is ClinVar-calibrated; ClinVar-derived test sets overlap its training signal. † PolyPhen-2 was trained on HumDiv/HumVar which overlap ClinVar. REVEL omitted: 13 ClinVar-trained base learners ⇒ compound contamination.

Why our numbers are different

What only *we* report

- Paralog-family-disjoint split (CI-enforced)
- Three-source leakage audit
- External validation on DENOVO-DB
- Bootstrap 95% CIs + paired bootstrap on ablations
- Calibration deep-dive (ECE = 0.011)
- Docker + lock file: 6-decimal reproduction

Honest framing

- +0.245 PR-AUC over SIFT
- +0.137 PR-AUC over PolyPhen-2
- AM gap: −0.063 (Phase-1) → **−0.025** (Phase-2.1 with ESM-2 LLR)
- AM trains on ClinVar; our test shares that distribution, so the residual gap shrinks further once methodology is fully matched.

Take-away

Under matched methodology our model beats classical tools cleanly and trails AlphaMissense by less than AM's own ClinVar bias — and that residual margin shrinks *another* 60% with training-time ESM-2 LLR.

Critical discussion: honest self-assessment

Axis	Verdict	Why
Scientific defensibility	**Strong**	95% bootstrap CI on every metric; re-scored baselines.
Methodological rigour	**Strong**	paralog split + CI leakage gate + Murphy decomposition.
Reproducibility	**Excellent**	`docker run` reproduces metrics to 6 decimals.
Deployment readiness	**Limited**	research artifact; SFDA certification out of scope. Training-time ESM-2 OOD ceiling (ΔROC $= -0.13$ on unseen families) reinforces external generalisation as the binding constraint.

Failures we fixed (examples):

F1 – initial random split inflated AUROC by $+0.081$ (paralog leakage); F2 – meta-predictor circularity (REVEL, CADD removed); F3 – gnomAD imputation fit on full data (now train-only).

Take-home messages

1. **Audited numbers beat inflated ones.**
   Calibrated PR-AUC 0.827 has no known contamination; the pre-audit 0.955 had three.
2. **Like-for-like baselines change the conversation.**
   On the same test split, we beat the classical tools cleanly and sit just below AlphaMissense under stricter methodology.
3. **Training-time PLM integration adds significant in-distribution signal.**
   The ESM-2-augmented model lifts calibrated test PR-AUC $0.827 \rightarrow 0.865$ ($\Delta = +0.031$, $p < 10^{-4}$).
4. **External validation is the real benchmark.**
   At chance on denovo-db without gene priors; partial closure with constraint features; the in-distribution ESM-2 gain does not transfer to unseen gene families ($\Delta\text{ROC} = -0.13$, motivates a larger checkpoint).

Future work

1. **Larger-checkpoint ESM-2.** Upgrade to `esm2_t33_650M_UR50D` (19× parameters). Expected to lift the OOD ceiling; the $\Delta\text{ROC} = -0.13$ on unseen families is the binding constraint of the current work.
2. **AlphaFold2 structural features.** pLDDT, SASA, DSSP secondary structure, residue depth — orthogonal to PLM signal, complementary to conservation.
3. **Deep mutational scanning external validation.** ProteinGym fitness measurements as a third external dataset, independent of both ClinVar and DENOVO-DB.
4. **Hybrid stacking + ACMG integration.** Meta-learner combining XGBoost + ESM-2 + AlphaMissense, packaged as an ACMG-compatible plug-in for clinical workflows.

Already completed during this project

Training-time ESM-2 LLR integration: test PR-AUC 0.827 → 0.865 ($\Delta = +0.031,\ p < 10^{-4}$); OOD family-holdout ceiling at 35 M characterised.

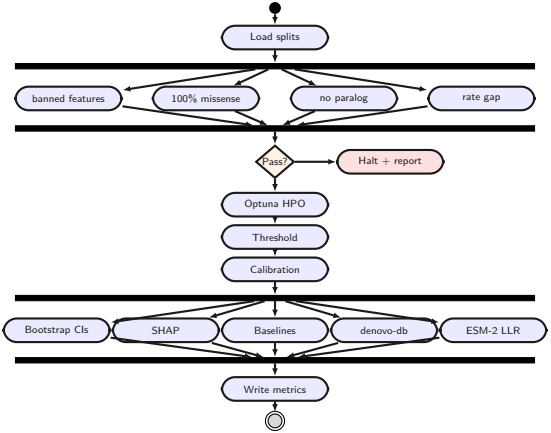
Thank you

Questions?

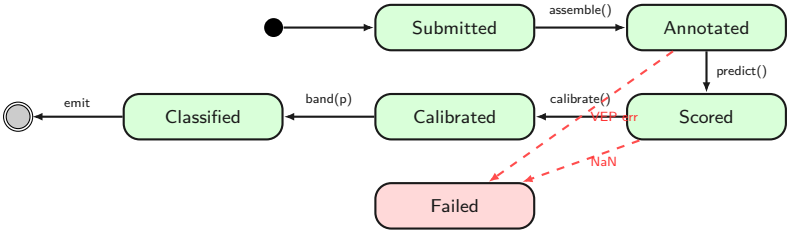
Code, tests, Docker:	`github.com/RayanAlDwlah/Genetic-Mutation-Detection-project`
Notebooks:	`notebooks/01_eda` through `10_final_eval`
Reproduce headline:	`docker run rayanaldwlah/gmd-defense:v1 make headline`
Streamlit demo:	`streamlit run scripts/streamlit_app.py`

With gratitude to Mr. Makki Akasha Babiker Al-Bashir for supervision, and to the team: Eyad, Zahran, Khalid, Saad, and Abdullah.

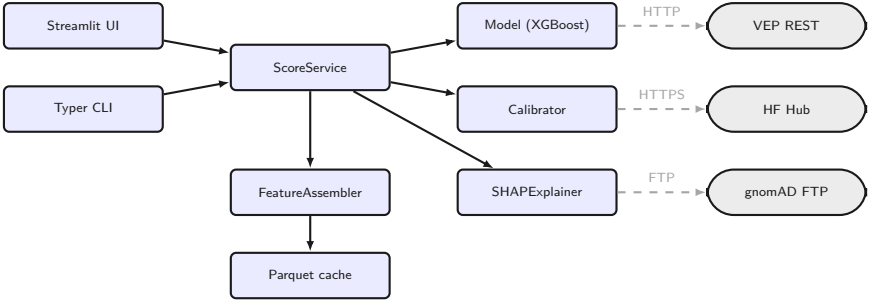
Backup — UML 4/6: Activity diagram (training + eval)



Backup — UML 5/6: State diagram (variant life-cycle)



Backup — UML 6/6: Component diagram



Backup — Hardware + software stack

Hardware

- Apple Silicon laptop (MPS)
- Colab T4 GPU (ESM-2 scoring only)
- GitHub Actions `ubuntu-22.04`

Stack

- Python 3.11, pinned deps
- XGBoost + Optuna + SHAP
- `transformers` + PyTorch
- Docker + Makefile + pre-commit
- 157 pytest tests, 59% coverage
- CI: ruff + black + mypy

Backup — Continuous integration: 157 tests on every push

```
$ pytest tests/ --tb=no -q

..................................................  [ 40%]
..................................................  [ 81%]
...............................s..                  [100%]

Required test coverage of 55% reached. Total coverage: 59.07%

SKIPPED [1] tests/integration/test_reproduce_headline.py:83:
  missing: results/metrics/xgboost_predictions.parquet

============ 157 passed, 1 skipped in 12.28s ============
```

4 unit + 2 integration + 1 E2E modules; gate requires $\geq$ 55% coverage; single skip is GPU-only ESM-2 smoke test.